Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

CHEMISTRY **0620/43**

Paper 4 Theory (Extended) **May/June 2017**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2017 series for most Cambridge IGCSE®, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

® IGCSE is a registered trademark.

This syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **8** printed pages.

© UCLES 2017

www.root19.com +91 - 9969 353 391

Question	Answer	Marks
1(a)(i)	**J**	**1**
1(a)(ii)	**E**	**1**
1(a)(iii)	**D**	**1**
1(a)(iv)	**J**	**1**
1(a)(v)	**L**	**1**
1(a)(vi)	**D**	**1**
1(b)(i)	(atoms with) same **number** of protons / atomic number / of same element	**1**
	different **numbe**r of neutrons / different mass number / different nucleon number	**1**
1(b)(ii)	**E AND G**	**1**
1(b)(iii)	they have the same number of electrons in their outer shell	**1**

ROOT 19

Question	Answer	Marks
2(a)(i)	CH_2	**1**
2(a)(ii)	initial colour: orange	**1**
	final colour: colourless / none	**1**
2(b)(i)	approximately horizontal line draw to right of and below the reagent line	**1**
	energy change shown starting level with the reactant energy **AND** finishing level with the product energy **AND** having only one (correct) arrow head **AND** labelled ΔH / energy change	**1**
2(b)(ii)	(energy required to break bonds =) 3624	**1**
	(energy given out when bonds made =) 3738	**1**
	–114 (kJ / mol)	**1**
2(c)	(attractive) forces between molecules	**1**
	(forces of attraction) are stronger in iodine	**1**

ROOT 19

www.root19.com +91 - 9969 353 391

Question	Answer	Marks
3(a)	regular arrangement / lattice of positive ions / magnesium ions / Mg^{2+} ions	**1**
	sea of electrons **OR** delocalised electrons	**1**
	attraction between (positive) ions and (delocalised / sea of) electrons	**1**
3(b)	electrons	**1**
	move / flow (throughout / through the structure)	**1**
3(c)	layers (of atoms or ions)	**1**
	layers / atoms / ions can slide / slip / glide (over each other) (without breaking the metallic bonds)	**1**
3(d)(i)	magnesium shown as (2, 8) using crosses	**1**
	sulfide shown as (2, 8, 8), with the two gained electrons in the outer shell of sulfur shown as crosses and all other electrons on sulfur shown as dots	**1**
	magnesium ion charge as 2^+ **AND** sulfide charge as 2^-	**1**
3(d)(ii)	melt / fused	**1**
	ions can move **OR** are mobile	**1**

ROOT 19

Question	Answer	Marks
4(a)	petroleum	**1**
4(b)(i)	*saturated:* only single bonds **OR** no double / multiple bonds (between carbon atoms)	**1**
	hydrocarbon: (compound that) contains carbon and hydrogen	**1**
	hydrocarbon: and no other elements / only	**1**
4(b)(ii)	alkane(s)	**1**
4(b)(iii)	any 2 from: • same / similar chemical properties • (same) general formula • (consecutive members) differ by CH_2 • same functional group • common (allow similar) methods of preparation • physical properties vary in predictable manner / show trends / gradually change / example of a physical property variation	**2**
4(b)(iv)	CO_2 and H_2O on right-hand side and no other products / reagents	**1**
	11 (O_2), 7 (CO_2), 8 (H_2O)	**1**
4(c)(i)	acid rain	**1**

© UCLES 2017

www.root19.com +91 - 9969 353 391

Question	Answer	Marks
4(c)(ii)	*carbon monoxide*: from incomplete combustion (of fuel)	**1**
	oxides of nitrogen: nitrogen (from the air) reacts with oxygen (from the air)	**1**
	oxides of nitrogen: at high temperatures (in engine) **OR** (electrical) spark (in the engine)	**1**
4(c)(iii)	poisonous / toxic / death	**1**
4(c)(iv)	any 3 from: • oxides of nitrogen are reduced / lose oxygen (to form nitrogen) • oxides of nitrogen form nitrogen • (oxides of nitrogen) react with carbon monoxide • gases (adsorb / stick) on the catalyst's surface	**3**
4(d)(i)	butane	**1**
4(d)(ii)	(molecules with) the same molecular formula	**1**
	different structural formula / different displayed formula	**1**
4(d)(iii)	**UV** light / **sun**light	**1**
	H–C*l*	**1**
	any mono to deca chloro-substituted derivative of methyl propane	**1**

ROOT 19

www.root19.com +91 - 9969 353 391

Question	Answer	Marks
5(a)(i)	loss (of electrons)	**1**
5(a)(ii)	$Ni \rightarrow Ni^{2+} + 2e^{-}$	**1**
5(a)(iii)	goes down / gets less / decreases / lower / smaller	**1**
5(b)(i)	beryllium	**1**
	most negative voltage with any (named) metal **OR** biggest voltage with cobalt / nickel	**1**
5(b)(ii)	cobalt **AND** nickel	**1**
5(b)(iii)	– sign	**1**
	2.7	**1**
5(c)	(set up cell) using magnesium and beryllium (electrodes)	**1**
	voltage positive if magnesium is metal 2	**1**
	OR	
	(set up cells) using both magnesium and beryllium with the same metal as the other electrode	**1**
	larger (magnitude) voltages with magnesium	**1**
	OR	
	use magnesium with a different metal and compare to a reference value in a table	**1**
	value is more negative than with beryllium, if magnesium is metal 1	**1**

© UCLES 2017
www.root19.com

+91 - 9969 353 391

Question	Answer	Marks
6(a)(i)	$BaCO_3 \rightarrow BaO + CO_2$	**1**
6(a)(ii)	anything pH in the range pH 10 to pH 14	**1**
6(a)(iii)	nitrogen dioxide	**1**
	oxygen	**1**
6(b)(i)	$Na_2CO_3 + Ba(NO_3)_2 \rightarrow BaCO_3 + 2NaNO_3$ **M1** formula of $NaNO_3$ **M2** equation fully correct	**2**
6(b)(ii)	filter	**1**
	wash (the residue) using water	**1**
	dry the residue between filter papers / in a warm place	**1**
6(c)(i)	M_r = 197	**1**
	(9.85 / 197 =) 0.05 (mol)	**1**
6(c)(ii)	0.05 (mol)	**1**
6(c)(iii)	(0.05 × 24) = 1.2 (dm_3)	**1**
6(c)(iv)	moles of HC*l* at the start = (250 / 1000 × 1.00) = 0.25	**1**
	moles HC*l* in excess = 0.25 – (2 × 0.05) = 0.15 (mol)	**1**

ROOT 19